

Aptimizer Validation Protocol

How every calculated output is checked against an independent benchmark, and how the two are compared.

Two different questions

Validation work fails most often because these two get merged. They need different evidence and they fail for different reasons.

Verification	Does the code implement the standard correctly?	Reproduce a published worked example. SP 16, SP 34, the explanatory handbook to IS 1893, and the IITK-GSDMA guidelines all carry solved problems with printed answers. If Aptimizer does not return the printed answer, the code is wrong. Cheap, fast, and belongs in your test suite.
Validation	Does the answer match a real building?	Compare against a real project you hold documents for: sanctioned drawings, the structural drawing, the geotech report, the priced BOQ, the fire NOC. This is what tells you the model of reality is right, not just the arithmetic.

Do verification first. There is no point comparing against a real building until the arithmetic reproduces a textbook.

Method: What to Test, On What, and How to Record It

The reference projects

One project is an anecdote. Use three, chosen to stress different parts of the engine, and get full documentation for at least one.

- **Project A, fully documented.** A real, built or sanctioned residential complex where you can obtain the sanctioned architectural drawings, the structural drawing, the geotechnical report, the priced BOQ and the fire NOC. This one project validates more modules than the other two combined. A friendly developer, a consultant you worked with, or your college's own campus buildings are all realistic sources.
- **Project B, awkward geometry.** An irregular, non-rectangular plot with a corner condition and more than one road edge. This is what breaks setback and packing logic.
- **Project C, edge of the rules.** A plot sized so that FAR, height or parking sits right at a threshold, where a small input change should flip a compliance result.

Tolerance, and why it varies

A single tolerance across every module would be dishonest. Geometry is arithmetic and should match almost exactly. Cost carries market variation. Soil parameters dominate foundation results far more than the formula does. State the band per module and justify it, and an examiner will take the whole exercise more seriously than if you claimed five percent everywhere.

Recording

One row per comparison, in a spreadsheet you can put in an appendix. Columns: module, quantity, Aptimizer value, benchmark value, benchmark source, absolute difference, percentage difference, within tolerance, note. Percentage difference against the benchmark, not against Aptimizer.

Record the failures. A validation table where everything passes reads as fabricated, and a discrepancy you investigated and explained is stronger evidence of engineering judgment than a clean sheet.

Stage 1: Site: Plot geometry, development controls and site intelligence

Module	Optimizer output	Independent benchmark method	What to compare	Tolerance
Plot & Setbacks	Plot area from polygon, per-edge setbacks, buildable envelope area, permissible height	Draw the same plot in AutoCAD or QGIS from the survey plan. Apply setbacks per the local building bye-laws by hand. Measure the resulting envelope with the AREA command.	Plot area, envelope area, each setback distance, permissible height	±1%
Plot & Setbacks	Setback minimums, height vs road width, open space vs height	Read the governing clause directly from the state or municipal building bye-laws for the project's actual location, and from NBC 2016 Part 3. Tabulate required vs applied.	Every required value, clause by clause	Exact match
GIS Intelligence	Terrain, elevation range, slope, flood risk, road and feature detection	Model the site in QGIS. Load a Cartosat-1 or SRTM 30 m DEM, run Slope and Aspect from the Raster Terrain Analysis toolbox, and clip to the plot. Digitise roads and water bodies from the same OSM extract Optimizer queries.	Min, max and mean elevation, mean and max slope, distance to nearest water body and road	±5% elevation; ±10% slope
GIS Intelligence	Sun path, sun events, annual insolation, solar potential	Run the same latitude, longitude and roof area through NREL PVWatts or PVsyst. For shading and sun path, build the massing in SketchUp and run a shadow study, or use Ladybug in Rhino-Grasshopper for hourly analysis.	Sunrise and sunset times on solstices, annual kWh/kWp, roof generation estimate	±10%
GIS Intelligence	Accessibility, site suitability score, buildability score	No external tool produces these directly: they are composite indices Optimizer defines. Validate the inputs against QGIS instead, then check the index is monotonic and that a hand-computed score from the documented weights reproduces the app's number.	Input values, then arithmetic reproduction of the index	Exact on the formula

Stage 2: Design: Layout, areas and parking

Module	Optimizer output	Independent benchmark method	What to compare	Tolerance
Apartment Planning	Carpet, built-up and super built-up area, ground coverage, FAR, FSI, density	Model the same building in Revit and read areas off a BIM schedule, or take them off the AutoCAD plan by hand. For a real project, compare against the area statement on the sanctioned drawing and the ULB sanction letter, which states permissible and achieved FAR.	Each area figure, FAR, FSI, ground coverage percentage, dwelling units per hectare	±2%
Apartment Planning	Generated floor plate: room sizes, aspect ratios, circulation	Measure a real typical floor plan from a sanctioned drawing. Compare room areas, minimum dimensions and the carpet-to-plate efficiency ratio. Your plancheck.py already does this against one built drawing; widen it to three.	Room area distribution, minimum room dimension, plate efficiency	Bands, not point values
Site Layout	Tower placement, count, spacing, road and amenity areas	Lay out the same plot by hand in AutoCAD following the bye-law setbacks and open space rules. A planner would do this on paper first. Compare achieved unit yield and confirm every tower footprint is inside the envelope.	Tower count, total built-up achieved, minimum tower spacing, containment	Containment exact; yield ±10%
Parking	Demand per tower, supply, ECS conversion, shared pool allocation	Compute demand by hand from the state bye-law parking table for the actual carpet-area bands. Compare against the parking statement on a sanctioned drawing of a real project of similar size.	ECS required, ECS provided, visitor and accessible bay counts	Exact on counts

Stage 3: Engineering: IS and NBC calculations

Module	Optimizer output	Independent benchmark method	What to compare	Tolerance
Structural Loads	Dead, live and wind loads; column axial loads; preliminary column sizes	Two layers. Verification: reproduce a solved example from SP 16, SP 34 or the IITK-GSDMA guidelines and check Optimizer returns the published answer. Validation: build the same frame in STAAD.Pro or ETABS and read column axial loads off the analysis.	Design wind pressure, factored axial load per column, column section	±5% vs software; exact vs example
Seismic	Zone factor, fundamental period, S_a/g , A_h , design base shear	Hand calculation is short and worth doing: $A_h = (Z/2)(I/R)(S_a/g)$, $V_b = A_h \times W$. Then run the same model in ETABS with the IS 1893 seismic definition and compare base shear. The explanatory handbook to IS 1893 carries worked examples.	T, S_a/g , A_h , V_b , storey shear distribution	±10% vs ETABS; ±2% vs hand calc
Foundation	Bearing capacity, footing size recommendation	Compute net safe bearing capacity by hand per IS 6403 from a real geotechnical investigation report. In practice the SBC comes from the geotech consultant, not from a formula alone, so use a real report as the reference.	Net SBC, footing plan area for a given column load	±15%
Mix Design	Target strength, w/c ratio, cement content, aggregate proportions	Work the IS 10262:2019 mix design procedure by hand for the same grade and exposure. Compare against an actual mix design report from an RMC supplier or a lab trial mix, which is what a site would hold.	Target mean strength, w/c, cement kg/m ³ , fine and coarse aggregate kg/m ³	±5%
Water Infrastructure	Population, demand in lpd, underground and overhead tank capacity	Hand calculation per IS 1172 and NBC Part 9: occupancy x per capita demand, then the one-third overhead, two-thirds underground split. Compare against the design basis report from a real project's plumbing consultant.	Population, total lpd, UG and OH tank volumes	±5%
Storm Water & RWH	Runoff, harvesting potential, recharge pit sizing	Rational method by hand: $Q = CIA$, with rainfall intensity taken from IMD data for the actual city rather than a generic figure. Compare pit sizing against NBC Part 9.	Runoff coefficient, peak runoff, annual harvest volume, pit volume	±10%
Fire Safety	Travel distance, stair width, fire lift, static water storage	Check each value directly against NBC 2016 Part 4 tables. For a real project, compare against the fire NOC drawing approved by the state fire service, which states these explicitly.	Every required value against the clause	Exact match
Accessibility	Ramp gradient, accessible bays, lift and door widths	Check against NBC 2016 Part 3 and the Harmonised Guidelines 2021 under the RPwD Act. This is a checklist audit, not a calculation.	Each provision, pass or fail	Exact match
Column Grid	Recommended bay spacing, member sizes	Compare against the structural grid on a real sanctioned structural drawing, and against a quick span-to-depth rule check. ETABS can confirm the sections carry the load.	Bay dimensions, beam depth, column size	Bands, not point values
Green Rating	Preliminary GRIHA or IGBC score	Fill in the actual GRIHA v2019 or IGBC Green Homes v3.0 scorecard by hand for the same project. Both are published criterion by criterion.	Points per criterion and total	Within one rating band
Embodied Carbon	kgCO ₂ e total and per sq m	Recompute from the same quantities using published emission factors: the ICE database (University of Bath) or CII factors for Indian cement and steel.	Total kgCO ₂ e, kgCO ₂ e per sq m built-up	±15%

Stage 4: Cost & Programme: Quantities, money and time

Module	Optimizer output	Independent benchmark method	What to compare	Tolerance
BOQ & Quantities	Concrete, steel, brickwork, formwork and finishes quantities	Manual take-off in Excel following IS 1200, the Indian method of measurement, from the same drawings. This is exactly how a quantity surveyor works. Cross-check with Revit schedules or CostX if you have access, then against the BOQ of a real project.	Each major quantity line, and steel in kg per cu m of concrete	±5% concrete; ±10% steel
Cost Estimation	Rates, item costs, wastage, preliminaries, overhead and profit, contingency, GST, total	Price the same BOQ off CPWD DSR 2023 with correction slips, or the state PWD Schedule of Rates, which is the legal basis for public works estimates in India. Compare cost per sq m against published market rates for the same city.	Rate per item, total cost, cost per sq m, cost per flat	±10%
Programme	Activity durations, critical path, project finish, float	Build the same activity list in Primavera P6 or MS Project with the same logic and durations, and compare. Also draw the network by hand for a 15-activity subset and verify the forward and backward pass manually.	Project duration, critical path membership, total float per activity	Exact on path; ±5% duration
Feasibility & ROI	Revenue, cash flow, NPV, IRR, break-even	Rebuild the cash flow in Excel and use XNPV and XIRR on the same monthly series. This is the standard method in development appraisal.	NPV, IRR, break-even month, peak funding requirement	±1%

Modules With No External Benchmark

Four things Aptimizer produces cannot be checked against an industry tool, because no industry tool produces them. Saying so plainly is better than inventing a comparison, and each still has a defensible way to be tested.

Vastu audit

Vastu is normative, not physical. There is no measurement that makes a sector placement right or wrong. Validate it as a rules engine instead: take a published Vastu text or a practitioner's checklist, encode ten specific rules, and confirm the audit flags exactly the violations a practitioner would flag on the same plan. Report it as rule conformance, never as accuracy.

Layout optimisation

There is no correct answer to compare against, only better and worse ones. Test it as an optimiser: confirm every output satisfies the hard constraints (containment, spacing, setbacks) on a hundred random plot polygons, then compare achieved unit yield against a planner's manual layout on the same plot. Constraint satisfaction is pass or fail; yield is a comparison.

Data reliability score

It measures your own confidence, so there is nothing external to check it against. Test it by injection: deliberately break an input, such as removing the road edge or setting a wrong plot area, and confirm the specific check fires and the score drops. A reliability score that stays green when you feed it a broken project is worse than no score.

Optimiser recommendations

A suggestion is not a measurement. Verify instead that the claimed saving is real: apply the recommendation, recompute, and confirm the new cost or quantity matches what the optimiser predicted. That is checkable, and it is the claim that matters.

Is This How It Is Actually Done?

Mostly yes, with two corrections worth making before you write this up.

- **Revit for quantities is real, but IS 1200 is the authority.** BIM schedules are widely used, and equally widely distrusted for billing, because a model measures what was modelled and not what the method of measurement says to count. In India the quantity that settles an argument is the one taken off per IS 1200. Use Revit as a cross-check and the manual IS 1200 take-off as the benchmark, not the other way round.
- **Rates come from a schedule of rates, not from a tool.** CPWD DSR, kept current with correction slips, or the state PWD SOR, is the legal basis for public works estimates. Any cost validation that does not reference one of these will not convince a civil examiner.
- **QGIS for site analysis is standard.** So is ETABS or STAAD.Pro for structure, Primavera P6 for programme, and Excel for development appraisal. Those mappings are sound.
- **Nobody validates a whole platform at once.** Real practice is module by module against a known reference, which is exactly what the tables above set out.

If you only have time for four

Ranked by how much credibility each buys per hour spent, and by how likely you are to be asked about it.

#	Comparison	Why this one
1	Areas, FAR and FSI against a sanctioned drawing	Everything downstream is derived from these. If areas are wrong, quantities, cost and programme are all wrong, and no other result means anything.
2	Seismic base shear against a hand calculation	Short enough to work on a whiteboard in a viva, and it is the calculation an examiner is most likely to ask you to derive.
3	BOQ against a manual IS 1200 take-off	This is the one where a thumb-rule platform is normally caught out, and yours is not a thumb-rule platform. Show the difference.
4	Cost against CPWD DSR	Turns a plausible number into a defensible one, and connects the platform to the document the industry actually prices from.

One caution on framing. Do not describe this as proving Aptimizer is accurate. Describe it as establishing the range over which its outputs agree with established methods, and naming where they do not. That claim is defensible, and it is the one an examiner will accept.